

Official Flickr8k Benchmark

8,000 Photographic Images | 40,000 Captions (5 caps/img)

Train Split (75%)

6,000 Images | 30,000 Captions

Loss Optimization & Tuning

Validation Split (12.5%)

1,000 Images | 5,000 Captions

Checkpoint Selection (Val MR)

Held-Out Test Split (12.5%)

1,000 Images | 5,000 Captions

Certified Zero-Leakage Eval

Visual Preprocessing & Feature Tokenization

1. Resize & Center Crop: 224×224 pixels, 3 channels
2. Normalization: ImageNet mean & std
3. Patch Partition: 16×16 patches $\rightarrow L_I = 196 + 1$ [CLS]
4. Frozen ViT-B/16: $\mathbf{X}_I^{(0)} \in \mathbb{R}^{B \times 197 \times 768}$

Textual Preprocessing & Tokenization

1. RoBERTa Byte-Pair Encoding (BPE)
2. Fixed Sequence Truncation/Padding: $L_T = 64$
3. Mask Generation: $\mathbf{M}_T \in \{0, 1\}^{B \times 64}$ (1=valid, 0=pad)
4. Frozen RoBERTa: $\mathbf{X}_T^{(0)} \in \mathbb{R}^{B \times 64 \times 768}$